

MIDUS Twin Screening Project Summary

The 998 twin pairs participating in the MIDUS Twin Screening Project represent the first national sample of twin pairs ascertained randomly via the telephone.

SAMPLE/STUDY DESIGN

Two research organizations, ICR/AUS Consultants and Bruskin Associates, were hired to do the initial household screenings. The ICR research group did 2 waves of 1,000 respondents each week, whereas Bruskin Associates did one wave of 1,000 respondents per week. Both groups asked the identical questions:

- Q1. "On another subject, do you have any twins in you or your spouse's immediate family where BOTH twins are still living? By immediate family, I mean you or your spouse (or partner), your or your spouse's (or partner's) parents, brothers or sisters, and your children."

Data was collected on whether the respondent was a twin, another family member was a twin, there were no twins in the family, or, in some cases (ICR/AUS Consultants only), the respondent's spouse (or partner) was a twin. It should be noted that during a pre-test period, we included more extended family members who were twins but cut back to the above description of family members once we found that we had difficulty getting interviews from the extended family members. On average, out of 1,000 households screened, about 14.8% of respondents said "yes" to the above question. Specifically, 2.3% of respondents were themselves a member of a twin pair and 12.5% of respondents had a twin pair in their family as defined above.¹ The latter group of respondents are referred to as informants (INF) throughout this document.

If a respondent answered yes to Q1, they were then asked the following:

- Q2. "The Harvard Medical School in Boston is conducting the first national study of twins in the United States.

(If R is twin): Would it be okay for the Harvard researchers to call you about this important study?

(If R is INF): Would it be okay for the Harvard researchers to call you to find out how to get in touch with the twins in your family?"

On average, about 60% of respondents answered "yes" to Q2. Thus, for every 1,000 households called, we received names and numbers for about 83 - 86 individuals.¹

¹These numbers were calculated from a sample of data received from the ICR/AUS and Bruskin Associates weekly household screenings and not from the entire study period.

At this point, our interviewing staff at the Institute for Social Research (ISR; affiliated with the University of Michigan) called those individuals who agreed to let the Harvard researchers contact them. When first placing a call, the interviewer knew only whether this was a twin or an informant, and the name and age of the person they were calling (i.e., the information supplied to us from ICR/AUS and Bruskin Associates). Interviewers were trained to follow a number of different scripts when making these calls depending on who they were calling and how we initially got their name (e.g., a twin contacted by ICR/Bruskin Associates; a relative of a twin pair; a twin whose name and number was given to us by an informant). Interviewers first established how many twin pairs were in the person's family and whether or not each pair was eligible to participate in this study. In some cases, interviewers gave the toll free number of the ISR telephone facility to either the twin or informant. This could be used by individuals wishing to verify the legitimacy of the study. In addition, if an informant was unwilling to give us the names and phone numbers of their family members, they (the informant) simply gave the toll free number directly to one or both members of the twin pair who could then contact us if they were interested in participating in the study. In other cases, a study brochure and tailored letter was sent to the informant or twins which described the study and answered the specific questions they may have had (e.g., regarding issues of confidentiality).

ELIGIBILITY CRITERIA:

Specific eligibility criteria for this study included:

1. The twin pair had to be related to the INF as described earlier (i.e., "you or your spouse (or partner), your or your spouse's (or partner's) parents, brothers or sisters, and your children";
2. The twins had to be between the ages of 25 and 74;
3. Both twins had to have a residential telephone number (i.e., no college dormitories, prisons, nursing homes, etc.);
4. Both twins had to live in the continental U.S.;
5. Both twins had to speak English;
6. Both twins had to be mentally and physically able to do the interview.

FINAL RESULT CODES:

Approximately 26% of first contacts (both TWIN and INF) resulted in a completed twin interview which means that both members of the twin pair were contacted by our interviewer, agreed to participate, and completed a short 10 minute twin screening questionnaire which was designed to collect information necessary for diagnosing whether a given pair was identical or fraternal. Specifically, 998 twin pairs agreed to participate in the study and completed the twin screening questionnaire. (NOTE: We attempted to contact 3,853 individuals who supplied

preliminary information to ICR or Bruskin Associates. Thus, 998 completed interviews/3,853 first contacts = 26%.)

Almost half of the first contacts (1,887/3,853 = 49%) identified twin pairs who were ineligible for the present study. The number of pairs corresponding to the different ineligibility codes are listed below.

Reason for Ineligibility	Number of Twin Pairs Ineligible
Wrong family relationship to INF	560 cases
Under 25 years of age	907 cases
Over 74 years of age	66 cases
Institution (One or both twins)	15 cases
Deceased (One or both twins)	104 cases
Not continental U.S. (One or both twins)	56 cases
Triplets/quadruplets	4 cases
No phone	57 cases
Exceeded age distribution requirements	70 cases
Physically or mentally unable to do Iw	28 cases
Do not speak English	5 cases
TOTAL # OF INELIGIBLE TWIN PAIRS	1,887 cases

NOTE: Of the 1,887 ineligible twin pairs, 498 pairs were ineligible for more than reason. In most cases (441 of the 498 pairs), the twins were both too young to participate and were of the wrong family relationship (e.g., young grandchildren of the informant).

Final result codes for the remaining names and numbers which did not lead to a completed interview are summarized on the next page.

Reasons for Incompleted Interviews	Number of Cases
Wrong information from ICR/Bruskin (No twins in family)	195 cases
INF or Twin1 refused to give information	176 cases
One or both twins refused to participate in the study	136 cases
INF did not know phone number of twins; Twin1 did not know phone number of Twin2	61 cases
Unable to reach contact person (INF or Twin) before the end of the study	394 cases
TOTAL # OF INCOMPLETED INTERVIEWS	968 cases

Thus, as stated earlier, 26% of the numbers supplied to us from ICR/AUS and Bruskin Associates led to a completed twin pair who agreed to participate in the study. It should be noted that the response rate varied substantially depending on whether the initial respondent from ICR/AUS or Bruskin Associates was a twin or an informant. Specifically, twin first contacts resulted in a completed twin pair 60% of the time (310 complete pairs/513 twin first contacts), whereas a completed twin pair resulted from an informant first contact 21% of the time (688 complete pairs/3,340 informant first contacts).

Once the twin screening questionnaire was completed by both members of a twin pair, their names and phone numbers were sent on to DataStat so that they could be recontacted by DataStat interviewers and complete the rest of the MIDUS study requirements. As expected, a number of pairs were lost at each stage of the study. This is particularly problematic in the case of twin studies as data from **both** members of the pair are necessary in order to perform any type of genetic analysis. (NOTE: When only one member of a twin pair completed the next step of the MIDUS, either the telephone interview or self-administered booklets, his or her data may be still be used to calculate overall mean scores, etc.) Complete data was received from both members of a twin pair for each stage of the project as follows:

Specific Stage of MIDUS Project	Number of Complete Twin Pairs
Twin screening questionnaire	998 twin pairs
MIDUS Telephone interview	924 twin pairs
Self-administered booklets	807 twin pairs
Cheek cell sample (DNA)	778 twin pairs

Twin pairs were diagnosed as identical or fraternal twins using self-report data collected as part of the initial twin screening questionnaire. Types of questions used in the diagnosis included whether the twins had the same physical features, whether their parents mistook them for each other when they were young, whether they perceived themselves to be almost exactly alike when young, and whether or not they themselves believed they were identical or fraternal twins. Specifically, two zygosity composites were calculated, one for each member of a pair using the following questions and point allocation.

Q13. When you and your twin were children, did the two of you look almost exactly alike or was there some difference in your appearance?

1. Looked alike

2. Some difference

Zygosity composite:

Respondents were given 2 points if they answered "looked alike", and 1 point if they answered "some difference."

Q14. Do you and your twin have the same eye color?

1. Yes

2. No

Q15. Do you and your twin have the same natural hair color?

1. Yes

2. No

Q16. Do you and your twin have the same complexion - that is, the same skin tone, color and texture?

1. Yes

2. No

Q17. Do you and your twin have the same facial features - that is, the same shape of eyes, nose, mouth, and face in general?

1. Yes

2. No

Zygosity composite:

Respondents were given 2 points if they answered "Yes" for questions #14 - #17. No points were given for "No" responses.

Q18. When you and your twin were children, would you say that you were generally the same height - that is, within an inch of each other - or was one of you generally taller than the other?

1. Same height 2. Respondent taller 3. Co-twin taller

Q19. Thinking again about when you were children, would you say that you generally weighed about the same - that is, within ten pounds of each other - or did one of you generally weigh more than the other?

1. Same weight 2. Respondent heavier 3. Co-twin heavier

Zygosity composite: Respondents were given 2 points if they checked response #1, "Same height/weight", for questions #18 and #19. No points were given for other responses.

Q20. When you were children, how often did your mother have difficulty telling you and your twin apart — all of the time, most of the time, some of the time, or never?

1. All of the time 2. Most of the time 3. Some of the time
4. Never 5. Inappropriate

Q21. When you were children, how often did your father have difficulty telling you and your twin apart — all of the time, most of the time, some of the time, or never?

1. All of the time 2. Most of the time 3. Some of the time
4. Never 5. Inappropriate

Zygosity composite: Respondents were given 4 points if they answered "All of the time", 3 points if they answered "Most of the time", 2 points if they answered "Some of the time", and 0 points for other responses.

Q29. In your opinion, are you and your twin definitely identical, probably identical, probably fraternal, definitely fraternal, or are you not sure what kind of twins you are?

- | | | |
|-------------------------|-----------------------|-----------------------|
| 1. Definitely identical | 3. Probably identical | 5. Probably fraternal |
| 7. Definitely fraternal | 9. Don't know | |

Zygosity composite:	Respondents were given 9 points if they answered "Definitely identical", 7 points if they answered "Probably identical", 3 points if they answered "Probably fraternal", 1 point if they answered "Definitely fraternal", and 0 points for other responses.
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Pairs were assigned to be identical (monozygotic, MZ) if one or both twins scored at least 18 on the zygosity composite, or if the sum of Twin 1 and Twin 2's scores exceeded 32 (i.e., in those cases, more weight was given to the twin who answered the questions consistent with an MZ diagnosis). The remaining pairs were diagnosed fraternal (dizygotic, DZ). In some cases, the zygosity of a twin pair was left as "undiagnosable", despite the twin zygosity composite scores, due to additional information found in the interviewer's marginal notes or, in most cases, because of one glaring inconsistency in responses. For example, in one pair, all data may have pointed toward a diagnosis of identical, yet one of the twins responded that the twins have different eye color, or that they currently differ in height by close to three inches. Analysis of DNA from the twins' cheek cell samples will resolve this issue in the future. In the meantime, some pairs are left as undiagnosable in the final twin file. NOTE: It is up to an individual's discretion as to whether or not s/he would like to "re-diagnose" these pairs and assume that error in the self-reported data will account for any inconsistencies between the twin responses in a given pair. It should also be noted that a small number of pairs had undergone testing or been told by a doctor at some point whether they were genetically identical or fraternal. This information was also taken into consideration when diagnosing zygosity, though it was not included when calculating the zygosity composite. Although similar methods of diagnosing zygosity have been shown to be over 90% accurate in diagnosing twin zygosity (e.g., Nichols and Bilbro, 1966), it is assumed that DNA analysis will be used to confirm the paper-and-pencil diagnosis in the future.

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